Midterm OperatorsActivity#2

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BSIT 2-3 OBJECT ORIENTED PROGRAMMING

General Instructions:

1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., System.out.println(86); without computation is
not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use System.out.println() for displaying output.
8. Screenshot the code and the output.
9. Compile and run your program to verify correctness before submission.
10. Submission Format: Section_LastName_FirstName_OperatorsActivity#2.pdf

Part I

Create three (3) separate Java programs following the required operators:
1. A program that uses bitwise operators.
2. A program that uses the ternary operator (? :).
3. A program that uses assignment operators (+=, -=, *=, /=).
Each program must produce the exact output required in the question and must compute the result using the specified operator.

Part II

Create one Java program that demonstrates the use of multiple operators.
Requirements:

1. Use at least five (5) different operators from the following categories:
o Arithmetic operators
o Relational operators
o Logical operators
o Unary operators

o Assignment operators
o Bitwise operators

2. The program must produce exactly six (6) outputs only.

3. At least one output must contain a tricky increment or decrement expression, such
as:

o a++ + ++a
o --b + b++
o ++x + x++
4. Do not reuse variable names, values, or logic patterns from Part I.
5. The outputs must include:
o At least two (2) boolean results
o At least four (4) numeric results

PART 1 – BITWISE

Bitwise.java | Ternary.java | Assignment.java

```java
public class Bitwise {
    public static void main(String[] args) {
        int a = 6; // 0110
        int b = 4; // 0100

        System.out.println("a & b = " + (a & b));
        System.out.println("a | b = " + (a | b));
        System.out.println("a ^ b = " + (a ^ b));
        System.out.println("~a = " + (~a));
    }
}
```

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<terminated> Bitwise (1) [Java Application] C:\Users\Ederlyn\.p2\pool\plugins\org.eclipse.justj.openjdk.hotsp

```
a & b = 4
a | b = 6
a ^ b = 2
~a = -7
```

PART 1 - TERNARY

Bitwise.java | Ternary.java | Assignment.java

```java
public class Ternary {
    public static void main(String[] args) {
        int num = 67;
        String result = (num % 2 == 0) ? "Even" : "Odd";
        System.out.println("Number is " + result);
    }
}
```

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```
<terminated> Ternary [Java Application] C:\Users\Ederlyn\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.2.v20260203-1141\jre\bin\javaw.exe  (Mar 13, 2026, 5:52:47 PM – 5:52:47 PM elapsed 0:0
Number is Odd
```

PART 1 - ASSIGNMENT

Bitwise.java | Ternary.java | Assignment.java

```java
public class Assignment {
    public static void main(String[] args){
        int x = 20;

        x += 10; //30
        System.out.println("x after += : " + x);

        x -= 25; //5
        System.out.println("x after -= : " + x);

        x *= 24; //120
        System.out.println("x after *= : " + x);

        x /= 4; //30
        System.out.println("x after /= : " + x);
    }
}
```

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```
<terminated> Assignment [Java Application] C:\Users\Ederlyn\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.2.v20260203-1141\jre\bin\javaw.exe  (Mar 13, 2026, 6:07:14 PM – 6:07:14 PM elapse
x after += : 30
x after -= : 5
x after *= : 120
x after /= : 30
```

PART 2 - MULTIOPERATORS

Bitwise.java | Ternary.java | Assignment.java | MultiOperators.java

```java
public class MultiOperators {
    public static void main(String[] args) {
        int x = 5, y = 15, z = 25;

        //Arithmetic
        System.out.println("x + y =" + (x + y));

        //Relational
        System.out.println("z > x = " + (z = x));

        //Logical
        System.out.println("(x < z) && (y < z ) = " + ((x < z) && (y < z)));

        //Unary with the joker trick
        int a = 6;
        System.out.println("x++ + ++x = " + (x++ + ++x));

        //Assignment
        int b = 10;
        b *= 2;
        System.out.println("y after *= : " + y);

        //Bitwise
        int u = 16, m = 19;
        System.out.println("u & m = " + (u & m));

    }
}
```

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<terminated> MultiOperators [Java Application] C:\Users\Ederlyn\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.2.v20260203-1141\jre\bin\javaw.exe (Mar 13, 2026, 6:25:00 PM – 6:25:00 PM ela

```
x + y =20
z > x = 5
(x < z) && (y < z ) = false
x++ + ++x = 12
y after *= : 15
u & m = 16
```